

MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

WAI4 - KGHS - KOREAN

Issue Date 05-Sep-2019

Revision Date 05-Sep-2019

Revision Number 1

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Name Nitrate 1 (as N)-LR

Product No AC4004-AMP

Pure substance/mixture Mixture

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent

Uses advised against No Information available

Supplier

Fisher Scientific Korea D5,D6, Incheon Airport Logistics Complex 150, Gonghangdong-Ro Mapo-Gu 296 Beon-Gil Jung-Gu, Incheon 400-340 Tel: +82-2-6196-5500 Fax: +82-2-6196-5501	Thermo Fisher Scientific© Water and Lab Products 22 Alpha Road Chelmsford, MA 01824, USA 1-978-232-6000
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E-mail address

Chem.KR@thermofisher.com info.water@thermo.com

Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970
CHEMTREC: 1-800-424-9300 or + 1-703-527-3887
Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

Made in USA

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

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Specific target organ toxicity - (single exposure)
Specific target organ toxicity - (repeated exposure)

Category 3
Category 2

Environmental hazards

Based on available data, the classification criteria are not met

Label Elements



Signal Word

Warning

Hazard Statements

H335 - May cause respiratory irritation

H336 - May cause drowsiness or dizziness

H373 - May cause damage to organs through prolonged or repeated exposure

Precautionary Statements

Prevention

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P271 - Use only outdoors or in a well-ventilated area

P261 - Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray

Response

P304 + P340 - IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing

P312 - Call a POISON CENTER or doctor/ physician if you feel unwell

P314 - Get medical advice/ attention if you feel unwell

Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P405 - Store locked up

Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

Other Hazards

NFPA

Health
2

Flammability
0

Instability
0

Physical hazards
N/A

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	Common Name	CAS-No	Index No.	Weight %
Water	No information available	7732-18-5	KE-35400	80 - 90
Ethylene Glycol	No information available	107-21-1	KE-13169	>=10 - 20
2-Propanol	No information available	67-63-0	KE-29363	2.5 - <5
Sulfanilic Acid	No information available	515-74-2	KE-01194	1.5 - 2

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	available			
2,5 Dihydroxybenzoic Acid	No information available	490-79-9	Not listed	1.5 - 2
Disodium 4,5-dihydroxynaphthalene-2,7-disulph onate	No information available	5808-22-0	Not listed	0.5 - 1
Cyclohexanedi-amine Tetraacetic Acid, Disodium Magnesium Salt	No information available	63451-33-2	Not listed	0.5 - 1
Proprietary Ingredients	No information available	999-99-9	Not listed	<0.1

SECTION 4: FIRST AID MEASURES

First aid measures

General Advice	If symptoms persist, call a physician.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.
Inhalation	Move to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.
Ingestion	Clean mouth with water and drink afterwards plenty of water.
Self-Protection of the First Aider	Use personal protective equipment. See section 8 for more information. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

Most important symptoms and effects, both acute and delayed

Most important symptoms and effects	None reasonably foreseeable
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Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically
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SECTION 5: FIREFIGHTING MEASURES

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media

No information available

Specific Hazards Arising from the Chemical

No information available.

Explosion Data

Sensitivity to Mechanical Impact None

Sensitivity to Static Discharge None

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full

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protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Personal Precautions

Ensure adequate ventilation. Use personal protective equipment.

Environmental Precautions

Should not be released into the environment. See Section 12 for additional ecological information.

Methods and Material for Containment and Cleaning Up

Methods for Containment

Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up

Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Handling

Wear personal protective equipment
Ensure adequate ventilation
Do not get in eyes, on skin, or on clothing
Avoid ingestion and inhalation

Conditions for Safe Storage, Including any Incompatibilities

Storage

Keep container tightly closed in a dry and well-ventilated place
Store at room temperature in the original container
Keep away from direct sunlight

Incompatible Products

No information available

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS-No	Korea	ACGIH TLV	OSHA PEL
Water	7732-18-5	Not listed	Not listed	Not listed
Ethylene Glycol	107-21-1	Ceiling: 100 mg/m ³	TWA: 25 ppm STEL: 50 ppm STEL: 10 mg/m ³	(Vacated) Ceiling: 50 ppm (Vacated) Ceiling: 125 mg/m ³
2-Propanol	67-63-0	STEL: 400 ppm TWA: 200 ppm	TWA: 200 ppm STEL: 400 ppm	(Vacated) TWA: 400 ppm (Vacated) TWA: 980 mg/m ³ (Vacated) STEL: 500 ppm (Vacated) STEL: 1225 mg/m ³ TWA: 400 ppm TWA: 980 mg/m ³
Sulfanilic Acid	515-74-2	Not listed	Not listed	Not listed
2,5 Dihydroxybenzoic Acid	490-79-9	Not listed	Not listed	Not listed
Disodium 4,5-dihydroxynaphthalene-2, 7-disulphonate	5808-22-0	Not listed	Not listed	Not listed
Cyclohexanedi-amine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not listed	Not listed	Not listed
Proprietary Ingredients	999-99-9	Not listed	Not listed	Not listed

Component	CAS-No	European Union	The United Kingdom	Germany
Water	7732-18-5	Not listed	Not listed	Not listed
Ethylene Glycol	107-21-1	TWA: 20 ppm (8h) TWA: 52 mg/m ³ (8h)	STEL: 40 ppm 15 min STEL: 104 mg/m ³ 15 min	TWA: 10 ppm (8 Stunden). AGW - exposure factor 2

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		STEL: 40 ppm (15min) STEL: 104 mg/m ³ (15min) Skin	STEL: 30 mg/m ³ 15 min TWA: 10 mg/m ³ 8 hr TWA: 20 ppm 8 hr TWA: 52 mg/m ³ 8 hr Skin	TWA: 26 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 10 ppm (8 Stunden). MAK can occur as vapor and aerosol at the same time TWA: 26 mg/m ³ (8 Stunden). MAK can occur as vapor and aerosol at the same time Höhepunkt: 20 ppm Höhepunkt: 52 mg/m ³ Haut
2-Propanol	67-63-0	Not listed	STEL: 500 ppm 15 min STEL: 1250 mg/m ³ 15 min TWA: 400 ppm 8 hr TWA: 999 mg/m ³ 8 hr	TWA: 200 ppm (8 Stunden). AGW - exposure factor 2 TWA: 500 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 200 ppm (8 Stunden). MAK TWA: 500 mg/m ³ (8 Stunden). MAK Höhepunkt: 400 ppm Höhepunkt: 1000 mg/m ³
Sulfanilic Acid	515-74-2	Not listed	Not listed	Not listed
2,5 Dihydroxybenzoic Acid	490-79-9	Not listed	Not listed	Not listed
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not listed	Not listed	Not listed
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not listed	Not listed	Not listed
Proprietary Ingredients	999-99-9	Not listed	Not listed	Not listed

Biological Exposure Indices

Component	CAS-No	Biological Exposure Indices
Water	7732-18-5	Not listed
Ethylene Glycol	107-21-1	Not listed
2-Propanol	67-63-0	40 mg/L Medium: urine Time: end of shift at end of workweek Determinant: Acetone
Sulfanilic Acid	515-74-2	Not listed
2,5 Dihydroxybenzoic Acid	490-79-9	Not listed
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not listed
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not listed
Proprietary Ingredients	999-99-9	Not listed

Appropriate engineering controls

Engineering Measures

Showers
Eyewash stations
Ventilation systems

Individual protection measures, such as personal protective equipment

Personal protective equipment

Use only those certified by the Korea Occupational Safety and Health Administration.

Eye/face Protection

Wear chemical splash goggles and face shield. If splashes are likely to occur, wear:
Face-shield.

Skin and Body Protection

Wear protective gloves/clothing.

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Respiratory Protection	None under normal use conditions. In case of inadequate ventilation wear respiratory protection.
Hygiene Measures	Handle in accordance with good industrial hygiene and safety practice.
Environmental exposure controls	No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical State	Liquid
Appearance	
Odor	Odorless
Odor Threshold	No information available
pH	3.5
PH Range	2-5

<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
Melting point/freezing point	No information available	
Boiling Point/Range	125 °C / 257 °F	
Flash Point (High in °C)	N/A	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit:	No information available	
Vapor pressure	No information available	
Vapor Density	No information available	
Specific Gravity	No information available	
Water Solubility	Soluble in water	
Solubility in other solvents	No information available	
Partition coefficient	No information available	
Autoignition Temperature	-	
Decomposition Temperature	No information available	
Kinematic viscosity	No information available	
Dynamic viscosity	No information available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

Other Information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No Information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY

Reactivity

No Information available

Chemical Stability

Stable under normal conditions

Possibility of Hazardous Reactions

None under normal processing

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Conditions to Avoid

Extremes of temperature and direct sunlight.

Incompatible Materials

No information available

Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 11: TOXICOLOGICAL INFORMATION

Product Information

Information on likely routes of exposure

Inhalation	Not an expected route of exposure.
Ingestion	May be harmful if swallowed.
Eyes	Avoid contact with eyes.
Skin	Avoid contact with skin.

Information on Toxicological Effects

Symptoms No information available

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Sensitization	No information available
Mutagenic Effects	No information available
Carcinogenicity	No information available.
Reproductive Effects	No information available
STOT - single exposure	No information available
STOT - repeated exposure	No information available
Aspiration hazard	No information available

Numerical measures of toxicity - Product Information

The following values are calculated based on chapter 3.1 of the GHS document .

ATEmix (oral)	4594 mg/kg
ATEmix (dermal)	59436 mg/kg
ATEmix (inhalation-dust/mist)	2420 mg/L

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity

3.9% of the mixture consists of components(s) of unknown hazards to the aquatic environment

Component	Freshwater Algae	Freshwater Fish	Water Flea
Ethylene Glycol 107-21-1	EC50: 6500 - 13000 mg/L, 96h (Pseudokirchneriella subcapitata)	LC50: = 41000 mg/L, 96h (Oncorhynchus mykiss) LC50: 14 - 18 mL/L, 96h static (Oncorhynchus mykiss) LC50: = 27540 mg/L, 96h static (Lepomis macrochirus)	EC50: = 46300 mg/L, 48h (Daphnia magna)

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		LC50: 40000 - 60000 mg/L, 96h static (Pimephales promelas) LC50: = 40761 mg/L, 96h static (Oncorhynchus mykiss) LC50: = 16000 mg/L, 96h static (Poecilia reticulata)	
2-Propanol 67-63-0	EC50: > 1000 mg/L, 72h (Desmodesmus subspicatus) EC50: > 1000 mg/L, 96h (Desmodesmus subspicatus)	LC50: = 11130 mg/L, 96h static (Pimephales promelas) LC50: > 1400000 µg/L, 96h (Lepomis macrochirus) LC50: = 9640 mg/L, 96h flow-through (Pimephales promelas)	EC50: = 13299 mg/L, 48h (Daphnia magna)
Sulfanilic Acid 515-74-2	EC50: = 91 mg/L, 72h (Desmodesmus subspicatus)	LC50: = 100.4 mg/L, 96h static (Pimephales promelas)	EC50: = 85.66 mg/L, 48h (Daphnia magna)

Persistence and Degradability

No information available

Bioaccumulation/ Accumulation

No information available

Mobility

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Component	log Pow
Ethylene Glycol 107-21-1	-1.93
2-Propanol 67-63-0	0.05

Other adverse effects

No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste Disposal Methods Disposal should be in accordance with applicable regional, national and local laws and regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport Not Regulated

IATA Not regulated

IMDG/IMO Not regulated
Marine Pollutant Not Applicable

Special Precautions for User No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

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Component	CAS-No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	AICS
Water	7732-18-5	KE-35400	X	231-791-2	X	X	-	X	-	X
Ethylene Glycol	107-21-1	KE-13169	X	203-473-3	X	X	-	X	X	X
2-Propanol	67-63-0	KE-29363	X	200-661-7	X	X	-	X	X	X
Sulfanilic Acid	515-74-2	KE-01194	X	208-208-5	X	X	-	X	X	X
2,5 Dihydroxybenzoic Acid	490-79-9	-	X	207-718-5	X	X	-	X	X	X
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	-	-	-	X	-	-	X	-	-
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	-	X	264-195-6	X	X	-	-	-	-
Proprietary Ingredients	999-99-9	-	-	-	-	-	-	-	-	-

Component	CAS-No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable
Ethylene Glycol	107-21-1	Not applicable	Not applicable	Not applicable	Not applicable
2-Propanol	67-63-0	Not applicable	Not applicable	Not applicable	Annex I - Y42
Sulfanilic Acid	515-74-2	Not applicable	Not applicable	Not applicable	Not applicable
2,5 Dihydroxybenzoic Acid	490-79-9	Not applicable	Not applicable	Not applicable	Not applicable
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not applicable	Not applicable	Not applicable	Not applicable
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not applicable	Not applicable	Not applicable	Not applicable
Proprietary Ingredients	999-99-9	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS-No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Water	7732-18-5	Listed	Not applicable	Not applicable
Ethylene Glycol	107-21-1	Listed	Not applicable	Not applicable
2-Propanol	67-63-0	Listed	Not applicable	Not applicable
Sulfanilic Acid	515-74-2	Listed	Not applicable	Not applicable
2,5 Dihydroxybenzoic Acid	490-79-9	Not applicable	Not applicable	Not applicable
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not applicable	Not applicable	Not applicable
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not applicable	Not applicable	Not applicable
Proprietary Ingredients	999-99-9	Not applicable	Not applicable	Not applicable

Korean National Regulations

Component	CAS-No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration
Water	7732-18-5	Annex 1 - KE-35400 Exempt (Index No. 25)	Not applicable	Not applicable
Ethylene Glycol	107-21-1	Annex 1 - KE-13169	Not applicable	Not applicable
2-Propanol	67-63-0	Annex 1 - KE-29363	Not applicable	Not applicable
Sulfanilic Acid	515-74-2	Annex 1 - KE-01194	Not applicable	Not applicable
2,5 Dihydroxybenzoic Acid	490-79-9	Not applicable	Not applicable	Not applicable
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not applicable	Not applicable	Not applicable
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not applicable	Not applicable	Not applicable
Proprietary Ingredients	999-99-9	Not applicable	Not applicable	Not applicable

Component	CAS-No	Chemical Control Act -	Chemical Control Act -	Chemical Control Act -
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		Toxic Chemicals	Prohibited Chemicals	Use Restricted Chemicals
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ethylene Glycol	107-21-1	Not applicable	Not applicable	Not applicable
2-Propanol	67-63-0	Not applicable	Not applicable	Not applicable
Sulfanilic Acid	515-74-2	Not applicable	Not applicable	Not applicable
2,5 Dihydroxybenzoic Acid	490-79-9	Not applicable	Not applicable	Not applicable
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not applicable	Not applicable	Not applicable
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not applicable	Not applicable	Not applicable
Proprietary Ingredients	999-99-9	Not applicable	Not applicable	Not applicable

Component	CAS-No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ethylene Glycol	107-21-1	Not applicable	Not applicable	Not applicable
2-Propanol	67-63-0	Not applicable	Not applicable	Not applicable
Sulfanilic Acid	515-74-2	Not applicable	Not applicable	Not applicable
2,5 Dihydroxybenzoic Acid	490-79-9	Not applicable	Not applicable	Not applicable
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not applicable	Not applicable	Not applicable
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not applicable	Not applicable	Not applicable
Proprietary Ingredients	999-99-9	Not applicable	Not applicable	Not applicable

Component	CAS-No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ethylene Glycol	107-21-1	Not applicable	Not applicable	Not applicable
2-Propanol	67-63-0	Not applicable	Not applicable	Not applicable
Sulfanilic Acid	515-74-2	Not applicable	Not applicable	Not applicable
2,5 Dihydroxybenzoic Acid	490-79-9	Not applicable	Not applicable	Not applicable
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not applicable	Not applicable	Not applicable
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not applicable	Not applicable	Not applicable
Proprietary Ingredients	999-99-9	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ethylene Glycol	107-21-1	Listed	Not applicable	Not applicable
2-Propanol	67-63-0	Listed	Not applicable	Not applicable
Sulfanilic Acid	515-74-2	Not applicable	Not applicable	Not applicable
2,5 Dihydroxybenzoic Acid	490-79-9	Not applicable	Not applicable	Not applicable
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not applicable	Not applicable	Not applicable
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not applicable	Not applicable	Not applicable
Proprietary Ingredients	999-99-9	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
Water	7732-18-5	Not applicable	Not applicable	Not applicable

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Ethylene Glycol	107-21-1	Listed	Listed	Not applicable
2-Propanol	67-63-0	Listed	Listed	Not applicable
Sulfanilic Acid	515-74-2	Not applicable	Not applicable	Not applicable
2,5 Dihydroxybenzoic Acid	490-79-9	Not applicable	Not applicable	Not applicable
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not applicable	Not applicable	Not applicable
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not applicable	Not applicable	Not applicable
Proprietary Ingredients	999-99-9	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ethylene Glycol	107-21-1	Not applicable	Ceiling: 100 mg/m ³	Not applicable
2-Propanol	67-63-0	Not applicable	STEL: 400 ppm TWA: 200 ppm	Not applicable
Sulfanilic Acid	515-74-2	Not applicable	Not applicable	Not applicable
2,5 Dihydroxybenzoic Acid	490-79-9	Not applicable	Not applicable	Not applicable
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not applicable	Not applicable	Not applicable
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not applicable	Not applicable	Not applicable
Proprietary Ingredients	999-99-9	Not applicable	Not applicable	Not applicable

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS-No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Ethylene Glycol	107-21-1	Not applicable	Not applicable	Not applicable	5. Group 3 Petroleum (Soluble) 4000 L	Not applicable	Not applicable
2-Propanol	67-63-0	Not applicable	Not applicable	Not applicable	3. Alcohol 400 L	Not applicable	Not applicable
Sulfanilic Acid	515-74-2	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
2,5 Dihydroxybenzoic Acid	490-79-9	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Proprietary Ingredients	999-99-9	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

Control Parameters

Component	CAS-No	Korea	Biological Exposure Indices
Water	7732-18-5	Not listed	Not listed
Ethylene Glycol	107-21-1	Ceiling: 100 mg/m ³	Not listed
2-Propanol	67-63-0	STEL: 400 ppm TWA: 200 ppm	40 mg/L Medium: urine Time: end of shift at end of workweek Determinant: Acetone
Sulfanilic Acid	515-74-2	Not listed	Not listed
2,5 Dihydroxybenzoic Acid	490-79-9	Not listed	Not listed

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Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not listed	Not listed
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not listed	Not listed
Proprietary Ingredients	999-99-9	Not listed	Not listed

US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

Component	CAS-No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Water	7732-18-5	Not applicable	Not applicable
Ethylene Glycol	107-21-1	Not applicable	Not applicable
2-Propanol	67-63-0	Not applicable	Not applicable
Sulfanilic Acid	515-74-2	Not applicable	Not applicable
2,5 Dihydroxybenzoic Acid	490-79-9	Not applicable	Not applicable
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not applicable	Not applicable
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not applicable	Not applicable
Proprietary Ingredients	999-99-9	Not applicable	Not applicable

CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

Component	CAS-No	CERCLA EHS RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ethylene Glycol	107-21-1	Not applicable	5000 lb	1.0
2-Propanol	67-63-0	Not applicable	Not applicable	1.0
Sulfanilic Acid	515-74-2	Not applicable	Not applicable	Not applicable
2,5 Dihydroxybenzoic Acid	490-79-9	Not applicable	Not applicable	Not applicable
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	5808-22-0	Not applicable	Not applicable	Not applicable
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	63451-33-2	Not applicable	Not applicable	Not applicable
Proprietary Ingredients	999-99-9	Not applicable	Not applicable	Not applicable

CLP Classification - Regulation (EC) No 1272/2008

Warning.

H335 - May cause respiratory irritation. H336 - May cause drowsiness or dizziness. H373 - May cause damage to organs through prolonged or repeated exposure.

P304 + P340 - IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing. P312 - Call a POISON CENTER or doctor/ physician if you feel unwell. P260 - Do not breathe dust/fume/gas/mist/vapors/spray.

SECTION 16: OTHER INFORMATION

Prepared By	Regulatory Affairs
Prepared For	Thermo Fisher Scientific Inc.©
Issue Date	05-Sep-2019
Revision Date	05-Sep-2019
Reason for revision	Initial Release.

SAFETY DATA SHEET

Nitrate 1 (as N)-LR

Revision Date 05-Sep-2019

MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

Disclaimer

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End of Safety Data Sheet